

HiRAG - Deep Semantic Understanding in RAG Workflows: Hands-on Tutorial



Fahd Mirza
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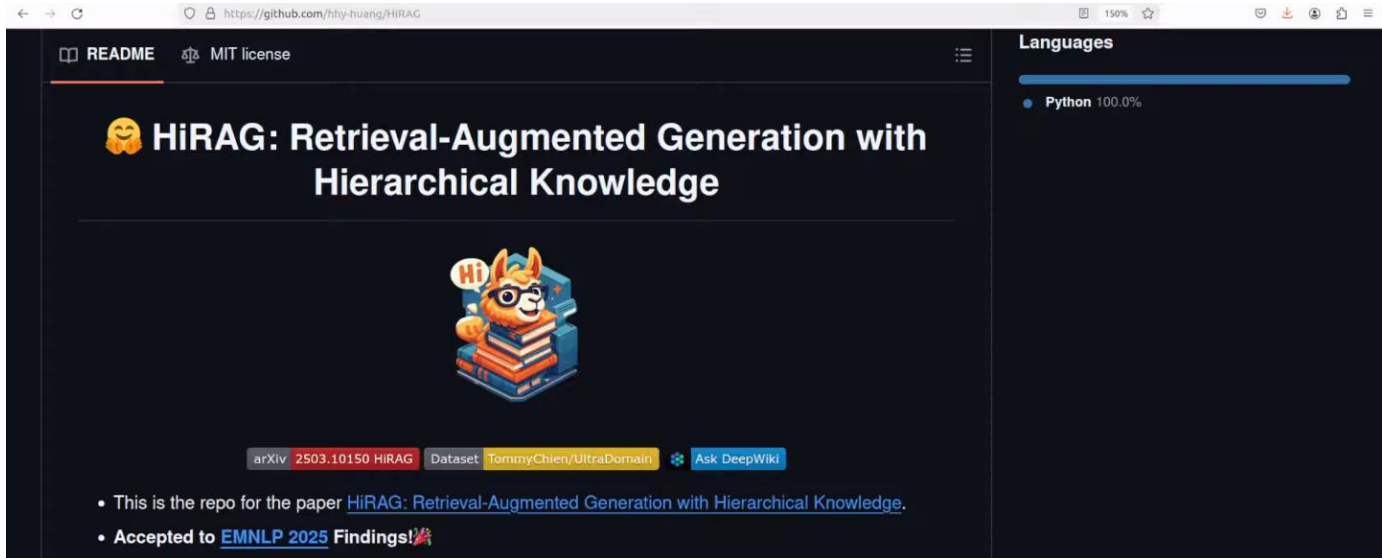
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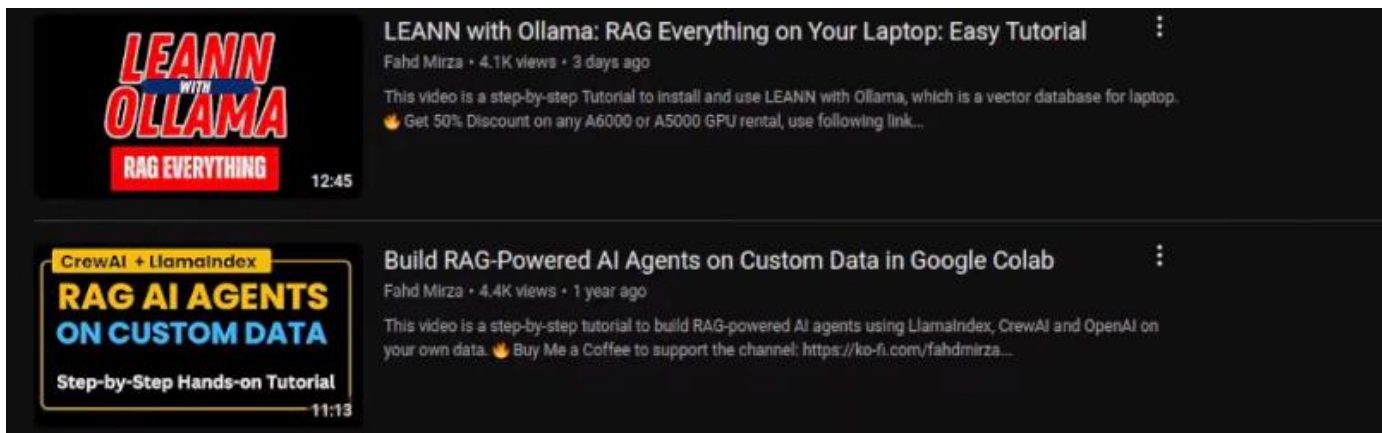
720 views Oct 15, 2025 #hirag

This video installs HiRAG, which utilizes hierarchical knowledge to enhance the semantic understanding and structure capturing capabilities of RAG systems in the indexing and retrieval processes.

This video is sponsored by <https://www.eigent.ai>, the World's First Multi-agent Workforce to Unlock Your Exceptional Productivity.



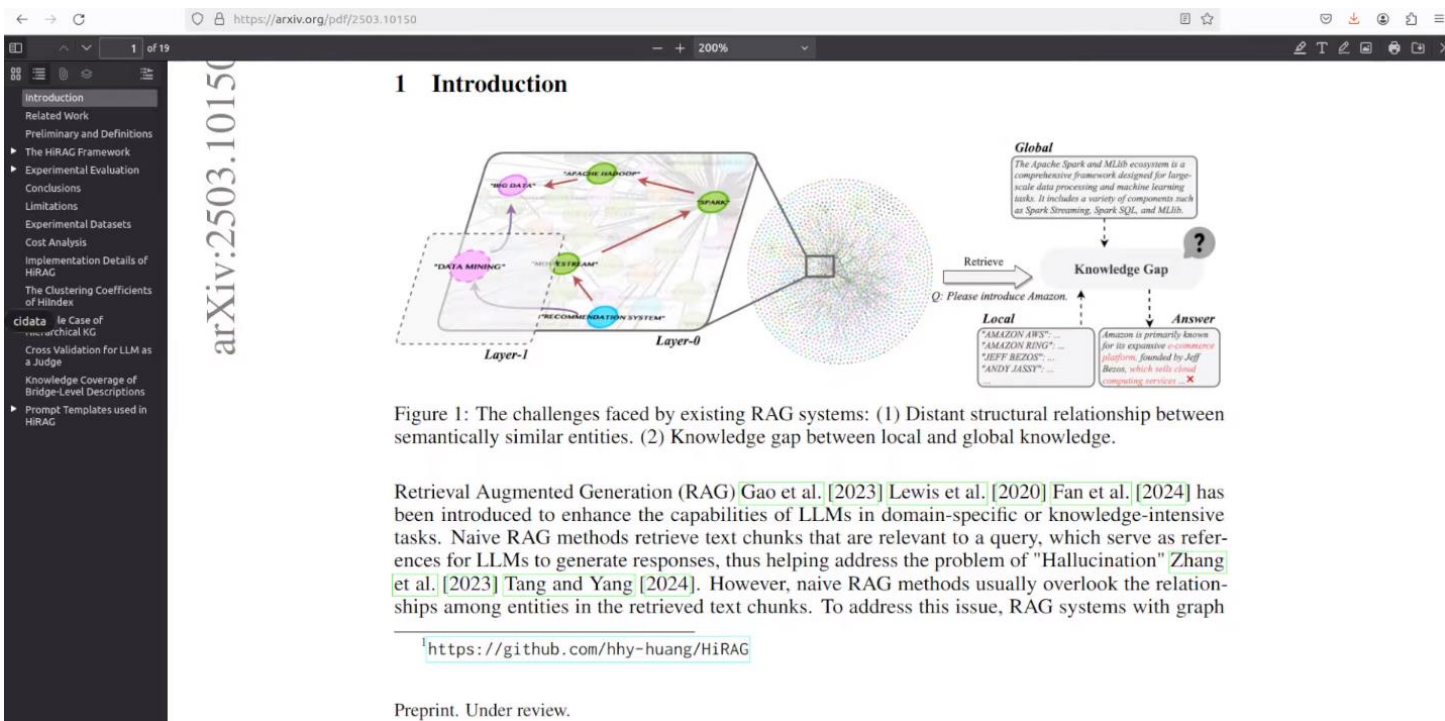
This only supports OpenAI models at present



```
Ubuntu@0360-kci2-ty6k-prmx100037: ~$ conda create -n ai python=3.11 -y && conda activate ai
Retrieving notices: ...working... done
Channels:
 - defaults
Platform: linux-64
Collecting package metadata (repodata.json): /
```

```
(ai) Ubuntu@0360-kci2-ty6k-prmx100037: ~$ git clone https://github.com/hhy-huang/HiRAG.git && cd HiRAG
Cloning into 'HiRAG'...
remote: Enumerating objects: 246, done.
remote: Counting objects: 100% (110/110), done.
remote: Compressing objects: 100% (56/56), done.
remote: Total 246 (delta 77), reused 55 (delta 54), pack-reused 136 (from 1)
Receiving objects: 100% (246/246), 28.17 MiB | 30.23 MiB/s, done.
Resolving deltas: 100% (115/115), done.
(ai) Ubuntu@0360-kci2-ty6k-prmx100037: ~/HiRAG$ pip install -e .
Obtaining file:///home/Ubuntu/HiRAG
Preparing metadata (setup.py) ... done
Collecting graspologic==3.4.1 (from HiRAG==0.1.0)
  Downloading graspologic-3.4.1-py3-none-any.whl.metadata (5.8 kB)
```

```
util-2.9.0.post0 python-dotenv-1.1.1 pytz-2025.2 pyyaml-6.0.3 ragas-0.2.2 regex-2025.9.18 requests-2.32.5 re
requests-toolbelt-1.0.0 safetensors-0.6.2 scikit-learn-1.6.1 scipy-1.12.0 seaborn-0.13.2 setuptools-73.0.1 six
-1.17.0 smart-open-7.3.1 sniffio-1.3.1 statsmodels-0.14.5 tenacity-9.0.0 threadpoolctl-3.6.0 tiktoken-0.7.0
tokenizers-0.21.4 tqdm-4.66.5 transformers-4.47.1 typing-extensions-4.15.0 typing-inspect-0.9.0 typing-inspe
ction-0.4.1 tzdata-2025.2 umap_learn-0.5.6 urllib3-2.5.0 wrapt-1.17.3 xxhash-3.5.0 yarl-1.20.1 zstandard-0.2
5.0
(ai) Ubuntu@0360-kci2-ty6k-prxm100037:~/HiRAG$ export OPENAI_API_KEY=
```



```
1 The Project Gutenberg eBook of Moby Dick; Or, The Whale
2
3 This ebook is for the use of anyone anywhere in the United States and
4 most other parts of the world at no cost and with almost no restrictions
5 whatsoever. You may copy it, give it away or re-use it under the terms
6 of the Project Gutenberg License included with this ebook or online
7 at www.gutenberg.org. If you are not located in the United States,
8 you will have to check the laws of the country where you are located
9 before using this eBook.
10
11 Title: Moby Dick; Or, The Whale
12
13 Author: Herman Melville
14
15 Release date: July 1, 2001 [eBook #2701]
16 Most recently updated: January 19, 2025
17
18 Language: English
19
20 Credits: Daniel Lazarus, Jonesey, and David Wldger
21
22 *** START OF THE PROJECT GUTENBERG EBOOK MOBY DICK; OR, THE WHALE ***
23
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28 MOBY-DICK;
29
30 or, THE WHALE.
31
32 By Herman Melville
33
34
35
36 CONTENTS
37
38 ETYMOLOGY.
39
40 EXTRACTS (Supplied by a Sub-Sub-Librarian).
41
42 CHAPTER 1. Loomings.
43
44 CHAPTER 2. The Carpet-Bag.
45
46 CHAPTER 3. The Spouter-Inn.
47
48 CHAPTER 4. The Counterpane.
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50 CHAPTER 5. Breakfast.
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EXPLORER
  HIRAG
    case
    docs
    eval
    hirag
    HIRAG.egg-info
    imgs
    .gitignore
    app.py
    ! config.yaml
    hi_Search_deepseek.py
    hi_Search_glm.py
    hi_Search_openai.py
    LICENSE
    readme.md
    requirements.txt
    setup.py

app.py
1 # app.py
2 from hirag import HiRAG, QueryParam
3
4 graph_func = HiRAG(
5     working_dir="/home/Ubuntu/testdata/",
6     enable_llm_cache=True,
7     enable_hierarchical_mode=True,
8     embedding_batch_num=6,
9     embedding_func_max_async=8,
10    enable_naive_rag=True
11)
12
13 with open("/home/Ubuntu/testdata/long.txt", "r") as f:
14     graph_func.insert(f.read())
15
16 print("Perform hi search:")
17 print(graph_func.query(
18     "Trace the reasoning from the Pequod's gams to Ahab's final decisions. Highlight the most influ
19     param=QueryParam(mode="hi")
20 ))
21
```

```
Ubuntu@0360-kci2-ty6k-prxm100037: ~/HIRAG
(ai) Ubuntu@0360-kci2-ty6k-prxm100037:~/HIRAG$ python app.py
2025-09-26 00:13:14,326 - Load KV full_docs with 0 data
2025-09-26 00:13:14,326 - Load KV text_chunks with 0 data
2025-09-26 00:13:14,326 - Load KV llm_response_cache with 0 data
2025-09-26 00:13:14,326 - Load KV community_reports with 0 data
2025-09-26 00:13:14,327 - Init {'embedding_dim': 1536, 'metric': 'cosine', 'storage_file': '/home/Ubuntu/testdata/vdb_entities.json'} 0 data
2025-09-26 00:13:14,327 - Init {'embedding_dim': 1536, 'metric': 'cosine', 'storage_file': '/home/Ubuntu/testdata/vdb_chunks.json'} 0 data
2025-09-26 00:13:14,337 - [New Docs] inserting 1 docs
2025-09-26 00:13:15,694 - [New Chunks] inserting 282 chunks
2025-09-26 00:13:15,694 - Insert chunks for naive RAG
2025-09-26 00:13:15,694 - Inserting 282 vectors to chunks
2025-09-26 00:13:17,974 - HTTP Request: POST https://api.openai.com/v1/embeddings "HTTP/1.1 200 OK"
2025-09-26 00:13:18,074 - HTTP Request: POST https://api.openai.com/v1/embeddings "HTTP/1.1 200 OK"
2025-09-26 00:13:18,122 - HTTP Request: POST https://api.openai.com/v1/embeddings "HTTP/1.1 200 OK"
2025-09-26 00:13:18,124 - HTTP Request: POST https://api.openai.com/v1/embeddings "HTTP/1.1 200 OK"
2025-09-26 00:13:18,134 - HTTP Request: POST https://api.openai.com/v1/embeddings "HTTP/1.1 200 OK"
2025-09-26 00:13:18,136 - HTTP Request: POST https://api.openai.com/v1/embeddings "HTTP/1.1 200 OK"
2025-09-26 00:13:18,551 - HTTP Request: POST https://api.openai.com/v1/embeddings "HTTP/1.1 200 OK"
2025-09-26 00:13:18,584 - HTTP Request: POST https://api.openai.com/v1/embeddings "HTTP/1.1 200 OK"
2025-09-26 00:13:18,708 - HTTP Request: POST https://api.openai.com/v1/embeddings "HTTP/1.1 200 OK"
2025-09-26 00:13:18,982 - HTTP Request: POST https://api.openai.com/v1/embeddings "HTTP/1.1 200 OK"
```

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Ubuntu@0360-kci2-ty6k-prxm100037: ~/HIRAG
2025-09-26 00:14:05,158 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:06,684 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:12,017 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:14,869 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:15,741 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:16,388 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:17,568 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:18,414 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:18,680 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:18,792 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:25,551 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:27,617 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:28,977 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:29,392 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:33,446 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:34,965 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:36,095 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:37,109 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:37,350 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:40,403 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:41,767 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:42,788 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
```

```
Ubuntu@0360-kci2-ty6k-prxm100037: ~/HIRAG
2025-09-26 00:14:53,251 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"
2025-09-26 00:14:53,279 - HTTP Request: POST https://api.openai.com/v1/chat/completions "HTTP/1.1 200 OK"

The gams (visits between whaleships) in Moby-Dick act as symbolic and narrative waypoints that reinforce Ahab's monomaniacal worldview. Early on, Ahab questions captains directly about Moby Dick; when they offer no useful intelligence or scoff—such as during the gam with the Bachelor, whose captain distrusts the whale's mythic aura—Ahab becomes more isolated and driven to rely solely on his internal compass rather than communal knowledge.
LitCharts
+2
Wikipedia
+2

In the encounter with the Jeroboam, for example, the ship's captain recounts losing a mate to Moby Dick and is accompanied by Gabriel, a prophetic figure. Gabriel equates the whiteness of the whale with divine power, predicting that Ahab is moving toward self-destruction. That encounter pushes Ahab deeper into the symbolic dimension of the whale—not just as prey but as cosmic nemesis.
Caitlin Duffy
+2
studylib.net
+2

The gam with the Samuel Enderby is another critical inflection: its captain, Boomer, has a prosthetic limb (
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Ubuntu@0360-kci2-ty6k-prxm100037: ~/HIRAG
+2
studylib.net
+2

The gam with the Samuel Enderby is another critical inflection: its captain, Boomer, has a prosthetic limb (like Ahab himself) caused by Moby Dick, yet bears no personal vendetta. He counsels caution, arguing that revenge may be futile. Ahab, however, refuses to be dissuaded. That contrast underscores the gulf—rational vs obsessive—that Ahab is deepening.
eNotes
+3
Wikipedia
+3
Caitlin Duffy
+3

Each gam thus contributes a data point: some deny the whale's significance, some warn of doom, some embody tempered acceptance. Ahab receives but rejects communal wisdom; instead he internalizes the whale's symbolism ever more tightly.

2. The Bridge: How Gams Feed into Intermediate Decisions

From these gams, certain encounters carry special weight, serving as "bridges" between mere encounters and Ahab's escalating resolve:
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2. The Bridge: How Gams Feed into Intermediate Decisions

From these gams, certain encounters carry special weight, serving as "bridges" between mere encounters and Ahab's escalating resolve:

Jeroboam / Gabriel's prophecy is a turning point: that encounter deepens Ahab's conviction that his quest is fated, spiritual, and preordained. It frames the whale not as an animal but as a cosmic adversary.

Samuel Enderby / Boomer's tempered stance is a counterpoint: since Boomer shares physical trauma (his lost limb) but has relinquished revenge, Ahab sees this as a challenge to suppress his own dissenting doubts and double down instead.

The gam with the Rachel is morally pivotal: Captain Gardiner begs Ahab to help find his lost son, lost in pursuit of Moby Dick. Ahab refuses, stating he "loses time"—this is the moment he refuses human compassion in favor of singular obsession.
CliffsNotes
+2
BookChat
+2
```



```
Ubuntu@0360-kci2-ty6k-prxmx100037: ~/HIRAG

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rsuit of Moby Dick. Ahab refuses, stating he "loses time"—this is the moment he refuses human compassion in
favor of singular obsession.
CliffsNotes
+2
BookChat
+2

These are the forks in his path: the prophetic doom, rational counterarguments, and a scene demanding a choi
ce between humanity and vengeance. Ahab time and again selects vengeance.

3. Final Decisions & the Chase – How the Chain Closes

By the time the final chase begins, Ahab's decisions are the sum total of his reactions to the gams:

He refuses to deviate for others (Rachel) or heed rational counsel (Enderby / Boomer).

He internalizes the whale as a cosmic sign (Jeroboam prophecy) and, having dismissed alternate viewpoints, a
ligns his entire voyage toward confrontational destiny.
```

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In the climactic three-day chase, every move—from which boats to lower, which routes to pursue, how to inter
pret Fedallah's prophecies—flows from that hardened conviction. In the end, Ahab's harpoon-line entangles hi
m; he is dragged under, fulfilling the very prophecies (the two hearses, only hemp kills) that originate in
earlier encounters.
Caitlin Duffy
+3
Wikipedia
+3
SparkNotes
+3

Thus the reasoning path is:

Gams present external voices and warnings →

Ahab filters or rejects them, favoring symbolic, inward logic →
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```
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earlier encounters.
Caitlin Duffy
+3
Wikipedia
+3
SparkNotes
+3

Thus the reasoning path is:

Gams present external voices and warnings →

Ahab filters or rejects them, favoring symbolic, inward logic →

That logic solidifies into fate, directing his final decisions

The chase executes those decisions, resulting in self-destruction.

(ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~/HIRAG$
```